

BIG SPRING, TX, USA

WMO: 720271

Lat: 32.213N

Lon: 101.521W

Elev: 2573

StdP: 13.38

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03044

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.9	24.7	1.4	6.5	36.7	6.2	8.3	40.0	31.6	53.3	27.4	52.6	7.1	20	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	100.6	67.1	99.5	67.1	97.5	68.2	73.4	87.6	72.4	86.5	71.5	85.4	11.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.4	123.2	76.9	69.5	119.4	76.2	68.2	113.9	76.2	38.7	87.2	37.8	86.2	37.0	85.5	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
26.7	24.0	20.8	DB	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			WB	15.2	106.5	4.7	5.8	11.8	110.6	9.0	114.0	6.4	117.3	2.9	121.5
				12.9	77.7	4.2	4.6	9.8	81.1	7.4	83.8	5.0	86.4	1.9	89.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.4	45.4	49.7	58.4	66.2	73.5	82.3	83.3	82.4	75.5	65.8	54.8	46.2	(6)
		DBStd	15.85	8.98	10.43	9.51	8.75	8.25	5.37	4.60	4.81	6.18	8.80	8.98	9.09	(7)
		HDD50	605	198	122	35	6	0	0	0	0	8	55	181		(8)
		HDD65	2401	609	432	244	90	24	0	0	0	4	96	319	583	(9)
		CDD50	6211	54	113	295	491	730	968	1031	1004	764	499	198	64	(10)
		CDD65	2532	0	3	39	125	288	518	566	539	318	122	13	1	(11)
		CDH74	30008	39	151	591	1571	3411	6336	6971	6321	3177	1199	200	41	(12)
		CDH80	15124	2	29	168	645	1724	3500	3829	3387	1396	409	33	2	(13)
(14)	Wind	WSAvg	10.2	9.3	10.4	11.6	12.5	11.8	9.7	8.7	8.6	9.2	9.2	9.2		(14)
(15)	Precipitation	PrecAvg	19.7	0.7	0.9	1.3	1.5	2.4	2.6	2.0	2.4	2.2	1.7	1.1	0.7	(15)
(16)		PrecMax	34.6	2.0	2.9	3.8	4.5	7.1	5.2	5.7	5.3	5.9	4.3	7.0	3.9	(16)
(17)		PrecMin	10.3	0.0	0.0	0.2	0.1	0.3	0.2	0.2	0.1	0.0	0.0	0.0		(17)
(18)		PrecStd	5.9	0.6	0.9	1.1	1.2	1.8	1.6	1.6	1.5	1.6	1.3	1.6	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	79.2	83.9	90.3	97.2	102.1	106.0	102.1	104.0	99.5	93.6	83.7	79.3		(19)
(20)		0.4% MCWB	53.7	53.9	57.1	59.8	62.6	66.1	68.0	67.5	66.7	65.1	58.6	53.2		(20)
(21)		2% DB	73.0	79.1	84.6	92.7	99.0	101.0	100.0	100.0	95.1	89.2	79.5	73.2		(21)
(22)		2% MCWB	50.9	53.4	56.7	59.6	63.0	67.5	68.4	68.2	67.2	65.2	57.1	52.9		(22)
(23)		5% DB	66.6	73.8	80.9	87.7	94.5	99.1	98.8	98.1	92.2	84.4	75.2	66.1		(23)
(24)		5% MCWB	48.6	52.5	55.7	58.3	63.1	67.4	68.8	68.8	67.4	63.0	56.8	50.2		(24)
(25)		10% DB	61.4	68.3	76.8	82.4	90.2	96.2	96.3	94.8	88.2	81.0	71.3	61.4		(25)
(26)		10% MCWB	46.6	50.1	55.7	58.1	63.5	68.4	69.2	68.5	66.6	62.5	54.6	47.8		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	59.2	60.6	65.9	68.0	71.6	74.5	74.5	74.3	74.3	71.4	64.4	60.5		(27)
(28)		0.4% MCDB	66.0	70.1	75.9	79.2	84.1	89.4	89.0	88.0	85.4	80.8	73.9	66.8		(28)
(29)		2% WB	54.0	57.8	63.2	66.2	70.4	73.0	73.0	73.0	72.1	69.6	62.1	56.4		(29)
(30)		2% MCDB	64.8	68.6	74.4	77.0	82.5	88.1	88.1	87.3	82.4	79.4	71.1	64.6		(30)
(31)		5% WB	51.2	54.9	60.7	64.3	69.1	71.8	72.0	72.0	70.9	68.1	59.8	53.1		(31)
(32)		5% MCDB	65.1	68.5	71.8	75.7	81.0	86.7	87.5	85.8	81.3	78.0	68.5	64.6		(32)
(33)		10% WB	48.0	52.1	58.5	62.3	67.8	70.9	71.1	71.1	69.6	66.3	57.4	50.0		(33)
(34)		10% MCDB	61.1	66.7	70.0	75.3	80.4	85.7	86.8	84.7	80.9	76.7	67.7	60.6		(34)
(35)	Mean Daily Temperature Range	MDBR	25.5	26.3	26.7	27.2	24.8	23.7	22.0	21.7	21.4	23.9	24.9	23.8		(35)
(36)		5% DB MCDBR	34.5	34.9	32.1	32.6	30.9	27.6	25.1	25.1	26.4	28.1	30.4	32.6		(36)
(37)		5% DB MCWBR	17.3	17.0	14.2	13.2	11.4	8.6	8.0	7.7	9.0	11.0	14.4	17.0		(37)
(38)		5% WB MCDBR	30.0	29.7	25.2	25.2	24.6	23.3	21.2	21.0	19.3	21.5	22.3	27.0		(38)
(39)		5% WB MCWBR	16.6	16.7	13.3	13.1	10.9	8.0	7.3	7.3	7.5	10.8	13.1	16.1		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.269	0.280	0.304	0.340	0.356	0.385	0.398	0.377	0.359	0.323	0.293	0.275		(40)
(41)		taud	2.579	2.554	2.498	2.391	2.417	2.375	2.395	2.424	2.429	2.516	2.534	2.563		(41)
(42)		Ebn at Noon	308	313	311	301	295	285	280	286	288	293	295	299		(42)
(43)		Edh at Noon	25	29	33	38	37	39	38	36	35	30	26	24		(43)
(44)	All-Sky Solar Radiation	RadAvg	1076	1324	1716	2111	2221	2377	2261	2096	1732	1411	1127	953		(44)
(45)		RadStd	109	152	125	135	182	121	133	99	123	174	123	92		(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+202	+140	(47)

Nomenclature: See separate page